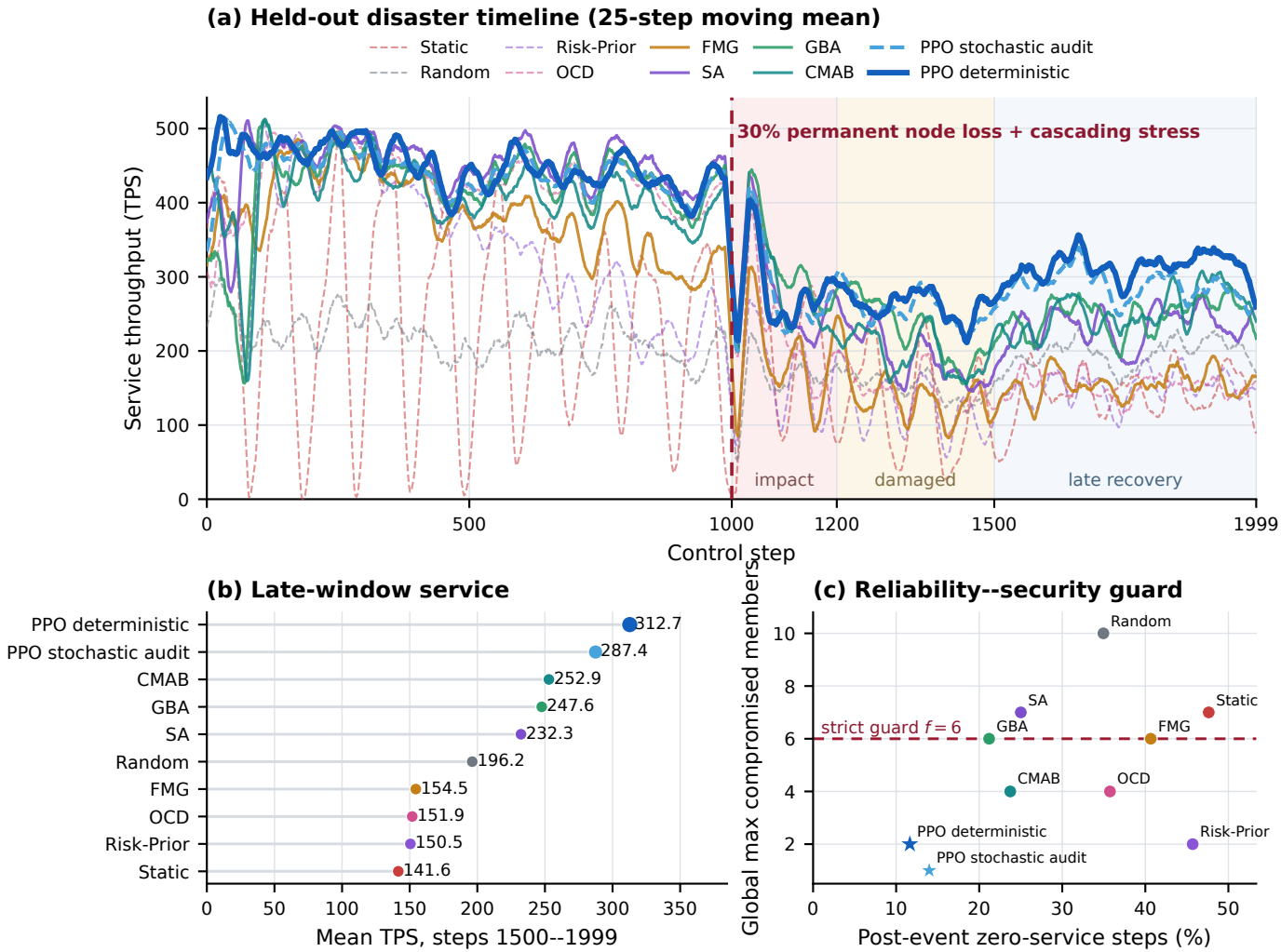

(a) Held-out disaster timeline (25-step moving mean)
Static
Random
Risk-Prior
OCD
FMG
SA
GBA
CMAB
PPO stochastic audit
PPO deterministic
30% permanent node loss + cascading stress
impact
damaged
late recovery
Service throughput (TPS)
Control step
(b) Late-window service
PPO deterministic
PPO stochastic audit
CMAB
GBA
SA
Random
FMG
OCD
Risk-Prior
Static
312.7
287.4
252.9
247.6
232.3
196.2
154.5
151.9
150.5
141.6
Mean TPS, steps 1500--1999
(c) Reliability--security guard
strict guard f = 6
Random
SA
Static
GBA
FMG
CMAB
OCD
PPO deterministic
Risk-Prior
PPO stochastic audit
Global max compromised members
Post-event zero-service steps (%)